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Microlab - Mini Modular Home Server Rack

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VIEW IN BROWSER

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Summary

Microlab: A fully 3D-printable, ultra-compact homelab for RPi, NUC, and more. Modular, customizable, and space-saving!

Gadgets > Computers

Tags: rack raspberrypi raspberrypi4 rackmount networking homelab miniab microlab raspberrypi5

Version 2 is here!

Meet the **Microlab**—a super compact, fully 3D-printable homelab that’s perfect for self-hosted tinkerers who love big ideas but small footprints. At just 5 inches wide and a tiny 160x160mm base, it’s even smaller than a 10" minilab, but don’t let its size fool you. With 4U or 6U height options (4U is around 260mm tall), this tiny rack is designed to fit your gear and your space.

The Microlab is all about flexibility. It’s modular, so you can mix and match panels, trays, and add-ons to create the perfect setup. Want to get creative? Customize your panels with cool infill patterns while slicing. Need cooling? Pop a 120mm fan on the top or bottom. And yes, it works both vertically and horizontally, so it’ll fit wherever you need it.

It’s built to handle your favorite small devices: **Raspberry Pi 4/5**, **TP-Link TL-SG105 Gigabit Switch**, **Intel NUC 13** (or other gens—they’re all pretty much the same size), **2.5" SATA HDD/SSD**, and more. Whether you’re running a budget x86 setup or a sleek ARM-based system, the Microlab keeps everything tidy and accessible.

The bolts are mainly **M3x8** with nuts from behind the columns. Normally, nuts should be pressed tight fit into the slots and stay there, if they don't stay in place you can use a little sticky tape to keep them in. Longer bolts than specified should be OK for most of the parts.

Perfect for homelabbers who want a clean, customizable, and space-saving solution, the Microlab is your go-to for building a powerful little lab that’s as unique as your projects.

Model files



cap.stl



col-4u.stl

col-6u.stl



panel-05u-solid.stl



panel-1u-solid.stl



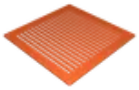
panel-2u-solid.stl



panel-4u-solid.stl



panel-cap-120mmfan.stl



panel-cap-slots.stl



panel-cap-solid.stl



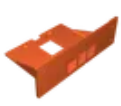
feet-edge.stl



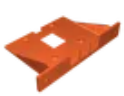
feet-round-10mm.stl



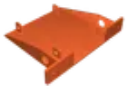
feet-round-15mm.stl



tray-1u-rpi5.stl



tray-05u-rpi4.stl



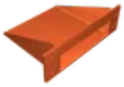
tray-05u-sata25.stl



tray-1u-nuc.stl



trayaddition-1-tlsg105.stl



tray-1u-tlsg105.stl

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